EVQ-Cosh: Theory-Only Notes (Theorems and Key Formulas)

paperdraft extract from `docs/paperdraft/CORE_THEORY.md`

2026-03-05

Abstract

This note collects the minimal theory required for a paper-ready EVQ-Cosh presentation: the exact broadband Euler–Lagrange ODE, its closed-form solution, the EVQ quantile warp (inverse CDF) in closed form, the $\tau \to 0$ geometric limit, the operator interpretation of the min kernel, and the Jensen "waterbed" inequality. Empirical claims are included only insofar as they justify the single approximation step (broadband projection).

1 Setup and Notation

Definition 1 (RoPE frequency map)**.** Fix a base $b > 1$. We parameterize frequencies by $\phi \in [0, 1]$ via

$$\omega(\phi) = b^{-\phi}. \tag{1}$$

Thus $\phi = 0$ corresponds to the highest frequency ($\omega = 1$) and $\phi = 1$ to the lowest frequency ($\omega = b^{-1}$).

Definition 2 (Channel density and discretization)**.** Let $\rho(\phi) \geq 0$ be a density on $[0, 1]$ with $\int_0^1 \rho(\phi)\,\mathrm{d}\phi = 1$. Given N discrete frequency channels, we use the standard RoPE indexing

$$u_k = \frac{k}{N}, \qquad k = 0, 1, \ldots, N-1, \tag{2}$$

and define discrete locations ϕ_k by the inverse CDF: $F(\phi_k) = u_k$, where $F(\phi) = \int_0^\phi \rho(t)\,\mathrm{d}t$. The RoPE inverse frequencies are $\omega_k = \omega(\phi_k) = b^{-\phi_k}$.

Remark 3 (Discrete endpoint convention). Under $u_k = k/N$, the discrete grid does not include $u = 1$ (hence typically $\phi_{N-1} < 1$). We use the continuous interval $[0, 1]$ for analysis and treat $\phi = 1$ as a normalized low-frequency boundary.

2 Exact Kernel and Broadband Projection

2.1 Exact kernel for the power-law prior

Definition 4 (Power-law distance prior)**.** For a training context length $L > 1$, define

$$D(\Delta) = \frac{1}{\Delta \ln L}\,\mathbf{1}_{\Delta \in [1, L]}. \tag{3}$$

Definition 5 (Phase-collision kernel)**.** The exact phase-collision kernel is the integral operator kernel

$$K(\phi_1, \phi_2) = \int_1^L D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\,\mathrm{d}\Delta. \tag{4}$$

Using $\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b))$ and the cosine integral $\mathrm{Ci}(x) = -\int_x^\infty \cos t/t \, \mathrm{d}t$, we obtain the closed form

$$K(\phi_1, \phi_2) = \frac{1}{2 \ln L} \Big[\mathrm{Ci}(\omega_- L) - \mathrm{Ci}(\omega_-) + \mathrm{Ci}(\omega_+ L) - \mathrm{Ci}(\omega_+) \Big], \tag{5}$$

where $\omega_- = |\omega(\phi_1) - \omega(\phi_2)|$ and $\omega_+ = \omega(\phi_1) + \omega(\phi_2)$.

2.2 Broadband projection (the only approximation)

Definition 6 (Broadband projection surrogate)**.** We approximate the exact kernel by the two-parameter family

$$K_{\mathrm{app}}(\phi_1, \phi_2) = \alpha \, \delta(\phi_1 - \phi_2) + \beta \, \min(\phi_1, \phi_2), \tag{6}$$

where $\alpha, \beta > 0$ are constants (fit/projection coefficients) and the residual is $\mathcal{E} = K - K_{\mathrm{app}}$.

Remark 7 (What is and is not claimed)*.* Equation (6) is the only approximation in the EVQ-Cosh derivation chain. In this project, (α, β) are treated as broadband effective coefficients (e.g. via Hilbert–Schmidt projection / finite-N calibration), and we do *not* claim the pointwise residual $\mathcal{E}$ is asymptotically small for large architectural bases (e.g. $b \approx 5 \times 10^5$). What is supported empirically is that the mid-band structure is captured with high fidelity (reported $R^2_{\mathrm{mid}} > 0.99$) and that the remaining matrix residual is dominated by boundary and diagonal-ridge artifacts.

3 Operator Viewpoint: min as a Green's Function

Definition 8 (Mixed-boundary Laplacian)**.** Let $A = -\frac{\mathrm{d}^2}{\mathrm{d}\phi^2}$ on $[0, 1]$ with mixed boundary conditions $g(0) = 0$ and $g'(1) = 0$. Its inverse A^{-1} is the integral operator with kernel $\min(\phi_1, \phi_2)$.

Lemma 9 (Green's function identity)**.** *For fixed $\psi \in (0, 1)$, the function $G(\phi, \psi) = \min(\phi, \psi)$ satisfies*

$$-\frac{\partial^2}{\partial \phi^2} G(\phi, \psi) = \delta(\phi - \psi), \qquad G(0, \psi) = 0, \qquad \frac{\partial}{\partial \phi} G(1, \psi) = 0, \tag{7}$$

in the sense of distributions.

Proof. For $\phi < \psi$, $G(\phi, \psi) = \phi$; for $\phi > \psi$, $G(\phi, \psi) = \psi$. Hence $\partial_\phi G$ jumps from 1 to 0 at $\phi = \psi$, so $\partial_\phi^2 G = -\delta(\phi - \psi)$. The boundary conditions follow from the piecewise form. □

Remark 10 (Surrogate operator family)*.* With this convention, the broadband surrogate $K_{\mathrm{app}} = \alpha I + \beta A^{-1}$ is the sum of an identity operator and a resolvent (Green's) operator. This is a convenient continuous model for the discrete diagonal ridge (I) plus smooth off-diagonal covariance (A^{-1}).

4 Variational Functional and Exact ODE Solution

4.1 Broadband functional

Consider the broadband functional (continuous density formulation)

$$\begin{aligned} \mathcal{J}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 \, \mathrm{d}\phi + \frac{\beta}{2} \int_0^1 \int_0^1 \rho(\phi_1) \rho(\phi_2) \min(\phi_1, \phi_2) \, \mathrm{d}\phi_1 \, \mathrm{d}\phi_2 \\ - \mu_F \int_0^1 \rho(\phi) b^{-2\phi} \, \mathrm{d}\phi + \lambda \left(\int_0^1 \rho(\phi) \, \mathrm{d}\phi - 1 \right). \end{aligned} \tag{8}$$

Here μ_F controls the strength of the Fisher-resolution term and λ enforces normalization.

Theorem 11 (Exact Euler–Lagrange ODE and general solution)**.** *Let $\rho^\star$ be a stationary point of (8). Then $\rho^\star$ satisfies the integral stationarity condition*

$$\alpha\rho(\phi) + \beta \int_0^1 \rho(\psi) \min(\phi, \psi)\ \mathrm{d}\psi - \mu_F b^{-2\phi} + \lambda = 0. \tag{9}$$

Moreover, ρ satisfies the (interior) non-homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma_F b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \qquad \gamma_F = \frac{\mu_F (2\ln b)^2}{\alpha}, \tag{10}$$

with boundary condition $\rho'(1) = -(2\mu_F \ln b/\alpha) b^{-2}$. For $\tau \neq 2\ln b$, the general solution is

$$\rho^\star(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + P b^{-2\phi}, \qquad P = \frac{\gamma_F}{4(\ln b)^2 - \tau^2}, \tag{11}$$

with constants C_1, C_2 determined by $\rho'(1)$ and $\int_0^1 \rho = 1$.

Proof. Taking the first variation of (8) yields Equation (9). Differentiating once in ϕ and using $\frac{\mathrm{d}}{\mathrm{d}\phi} \int_0^1 \rho(\psi) \min(\phi, \psi)\, \mathrm{d}\psi = \int_\phi^1 \rho(\psi)\, \mathrm{d}\psi$ gives

$$\alpha\rho'(\phi) + \beta \int_\phi^1 \rho(\psi)\ \mathrm{d}\psi + 2\mu_F \ln b\ b^{-2\phi} = 0. \tag{12}$$

Evaluating at $\phi = 1$ yields the boundary condition on $\rho'(1)$. Differentiating again and using $\frac{\mathrm{d}}{\mathrm{d}\phi} \int_\phi^1 \rho(\psi)\, \mathrm{d}\psi = -\rho(\phi)$ gives $\alpha\rho''(\phi) - \beta\rho(\phi) = \mu_F (2\ln b)^2 b^{-2\phi}$, i.e. (10). The closed-form solution (11) follows from constant-coefficient ODE theory; a particular solution $P b^{-2\phi}$ exists whenever $\tau \neq 2\ln b$. □

4.2 The cosh tether as the dominant broadband component

Remark 12 (Dominant cosh component). In practice, $\tau = \mathcal{O}(1)$ while $2\ln b \approx 20$–30 for common architectural bases, so the Fisher pulse $P b^{-2\phi}$ is sharply localized near $\phi = 0$. The macroscopic interference-suppression shape is therefore captured by the homogeneous "cosh tether".

Corollary 13 (Pure tether model yields an exact cosh density)**.** *Consider the simplified (Tikhonov + Brownian) functional with Fisher term removed:*

$$\mathcal{J}_{\alpha,\beta}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2\ \mathrm{d}\phi + \frac{\beta}{2} \int_0^1 \int_0^1 \rho(\phi_1)\rho(\phi_2) \min(\phi_1, \phi_2)\ \mathrm{d}\phi_1\, \mathrm{d}\phi_2 + \lambda \left(\int_0^1 \rho(\phi)\ \mathrm{d}\phi - 1 \right). \tag{13}$$

Then the unique minimizer satisfies

$$\rho^\star_{\alpha,\beta}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{14}$$

Proof. The stationarity equation is $\alpha\rho(\phi) + \beta \int_0^1 \rho(\psi) \min(\phi, \psi)\, \mathrm{d}\psi + \lambda = 0$. Differentiating twice as in the theorem proof yields $\alpha\rho''(\phi) - \beta\rho(\phi) = 0$ with boundary condition $\rho'(1) = 0$. Writing $\tau = \sqrt{\beta/\alpha}$, the general solution is $\rho(\phi) = A\cosh(\tau(1-\phi)) + B\sinh(\tau(1-\phi))$; the boundary condition forces $B = 0$. Normalization fixes $A = \tau/\sinh\tau$. □

5 Exact Variational Quantization (EVQ): Closed-Form Warp

Theorem 14 (CDF and inverse CDF for the cosh density)**.** *Let* $\rho_\tau(\phi) = \frac{\tau\cosh(\tau(1-\phi))}{\sinh\tau}$ *for* $\tau > 0$*. Then its CDF is*

$$F_\tau(\phi) = \int_0^\phi \rho_\tau(t)\,\mathrm{d}t = 1 - \frac{\sinh(\tau(1-\phi))}{\sinh\tau}. \tag{15}$$

For $u \in (0,1)$*, the inverse CDF is*

$$F_\tau^{-1}(u) = \phi(u;\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}\big((1-u)\sinh\tau\big). \tag{16}$$

Proof. Integrate $\cosh(\tau(1-\phi))$ explicitly: $\int_0^\phi \cosh(\tau(1-t))\,\mathrm{d}t = \frac{1}{\tau}(\sinh\tau - \sinh(\tau(1-\phi)))$. Multiplying by $\tau/\sinh\tau$ yields (15). Inverting $1-u = \sinh(\tau(1-\phi))/\sinh\tau$ gives (16). □

Corollary 15 (Geometric is the $\tau \to 0$ limit)**.** *Let* $u_k = k/N$*. Then* $\phi_k(\tau) = \phi(u_k;\tau)$ *satisfies* $\lim_{\tau\to 0}\phi_k(\tau) = u_k$*. Equivalently, EVQ degrades to geometric RoPE as* $\tau \to 0$*.*

Proof. For small τ, $\sinh\tau = \tau + \mathcal{O}(\tau^3)$ and $\operatorname{arcsinh}(x) = x + \mathcal{O}(x^3)$. Hence

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\tau + \mathcal{O}(\tau^3)) = 1 - \frac{(1-u_k)\tau + \mathcal{O}(\tau^3)}{\tau} = u_k + \mathcal{O}(\tau^2).$$

□

Proposition 16 (Endpoint anchoring)**.** *For any* $\tau > 0$*, the EVQ warp satisfies* $\phi(0;\tau) = 0$ *and* $\phi(1;\tau) = 1$*. In particular, under the discrete choice* $u_0 = 0$ *we have* $\phi_0 = 0$ *and* $\omega_0 = b^{-\phi_0} = 1$ *exactly, matching the geometric highest-frequency channel.*

Proof. Immediate from (16): $\operatorname{arcsinh}(\sinh\tau) = \tau$ and $\operatorname{arcsinh}(0) = 0$. □

Remark 17 (Local expansion (checked))*.* Expanding (16) at $\tau = 0$ gives

$$\phi(u;\tau) = u - \frac{u(1-u)(2-u)}{6}\tau^2 + \mathcal{O}(\tau^4), \tag{17}$$

so for $\tau > 0$ we have $\phi(u;\tau) < u$ for $u \in (0,1)$ (a shift toward the high-frequency end), while the *spacing* is reallocated asymmetrically: low-frequency gaps increase and high-frequency gaps shrink.

6 Waterbed Inequality (Integrated Trade-off)

Proposition 18 (Jensen waterbed bound)**.** *Define local Fisher information* $\mathcal{I}(\phi) = c\,\rho(\phi)\,b^{-2\phi}$ *for some constant* $c > 0$*, and assume a local error proxy satisfies* $E(\phi) \geq 1/\mathcal{I}(\phi)$*. Then*

$$\int_0^1 \ln E(\phi)\ \mathrm{d}\phi \geq \ln b - \ln c. \tag{18}$$

Moreover, if $E(\phi) = 1/\mathcal{I}(\phi)$ *and* ρ *is unconstrained beyond normalization, equality holds if and only if* ρ *is constant (geometric).*

Proof. By $E \geq 1/\mathcal{I}$, $\ln E \geq -\ln\mathcal{I}$. Hence

$$\int_0^1 \ln E\ \mathrm{d}\phi \geq -\int_0^1 \ln c\ \mathrm{d}\phi - \int_0^1 \ln\rho(\phi)\,\mathrm{d}\phi + 2\ln b\int_0^1 \phi\ \mathrm{d}\phi.$$

Since $\int_0^1 \phi\ \mathrm{d}\phi = \frac{1}{2}$, this is $\int_0^1 \ln E \geq -\ln c - \int_0^1 \ln\rho + \ln b$. Finally, Jensen's inequality for the convex function $-\ln x$ gives $-\int_0^1 \ln\rho(\phi)\,\mathrm{d}\phi \geq -\ln\int_0^1 \rho(\phi)\,\mathrm{d}\phi = 0$, with equality iff ρ is constant. □

7 Scaling Law for $\tau^\star$ (Status: Conjecture)

Remark 19 (What is proved vs. what is fit). The project contains strong empirical evidence that the optimal EVQ parameter $\tau^\star$ scales approximately as $d_{\text{head}}/\sqrt{L}$ at fixed base, but a fully rigorous derivation of the $\alpha(L, b)$ calibration is still an open proof task. We therefore record the scaling as a conjecture.

Proposition 20 (Single-variable scaling law (conjectural statement))**.** *Fix b and d_{head}. For a training length L, the optimal EVQ parameter obeys*

$$\tau^\star(L) \approx \frac{d_{\text{head}}}{\sqrt{L}}. \tag{19}$$

8 Summary: Copy-Paste Formula Sheet

EVQ warp (inverse CDF). For $N = d_{\text{head}}/2$ and $u_k = k/N$:

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}\big((1-u_k)\sinh\tau\big), \qquad \omega_k = b^{-\phi_k}. \tag{20}$$

Cosh density.

$$\rho_\tau(\phi) = \frac{\tau\cosh(\tau(1-\phi))}{\sinh\tau}, \qquad F_\tau(\phi) = 1 - \frac{\sinh(\tau(1-\phi))}{\sinh\tau}. \tag{21}$$

Broadband ODE and general solution.

$$\rho'' - \tau^2\rho = \gamma_F b^{-2\phi}, \qquad \rho(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + \frac{\gamma_F}{4(\ln b)^2 - \tau^2} b^{-2\phi}. \tag{22}$$

Waterbed inequality.

$$\int_0^1 \ln E(\phi)\ \mathrm{d}\phi \geq \ln b - \ln c. \tag{23}$$